

Name: _____

Class: _____

YEAR 12
ASSESSMENT TEST
TERM 1, 2018

Mathematics Extension 2

General Instructions

- Working time – 120 minutes + 5 minutes reading
- Board approved calculators may be used
- Write using black or blue pen
- All necessary working should be shown in all questions
- Write your student number at the top of every page of your answer sheets.

5 Multiple Choice Questions-1 mark each

5 Long Response Questions-15 marks each

Total marks – 80

Attempt all questions

Start each question on a new sheet of paper.

Multiple Choice Questions

Answer the following questions in the answer sheet provided

- 1) What is the multiplicity of the root $x = -1$ of the equation $3x^5 - 5x^4 - 35x - 27 = 0$?

(A) one

(B) two

(C) three

(D) four

- 2) Using a suitable substitution, the definite integral $\int_0^{\frac{\pi}{24}} \tan 2x \sec^2 2x \, dx$ is equivalent to

(A) $\int_0^{\frac{\pi}{24}} \frac{u}{2} \, du$

(B) $\int_0^{2-\sqrt{3}} 2u \, du$

(C) $\int_0^{\frac{\pi}{24}} 2u \, du$

(D) $\int_0^{2-\sqrt{3}} \frac{u}{2} \, du$

- 3) Given the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is e , then the eccentricity of the ellipse $\frac{x^2}{a^2+b^2} + \frac{y^2}{b^2} = 1$ is

(A) $\sqrt{e}$

(B) $\frac{1}{e}$

(C) e

(D) e^2

4) For a certain function $y = f(x)$, the function $y = f(|x|)$ is represented by

- (A) A reflection of $y = f(x)$ in the y axis
- (B) A reflection of $y = f(x)$ in the x axis
- (C) A reflection of $y = f(x)$ in the x axis for $y \geq 0$
- (D) A reflection of $y = f(x)$ in the y axis for $x \geq 0$

5) Let w be the complex root of unity such that $w^n = 1$, $w \neq 1$. Find the value of $\sum_{k=0}^{n-1} \left(w^k + \frac{1}{w^k} \right)$ where $n \geq 3$

- (A) 0
- (B) 1
- (C) 2
- (D) 3

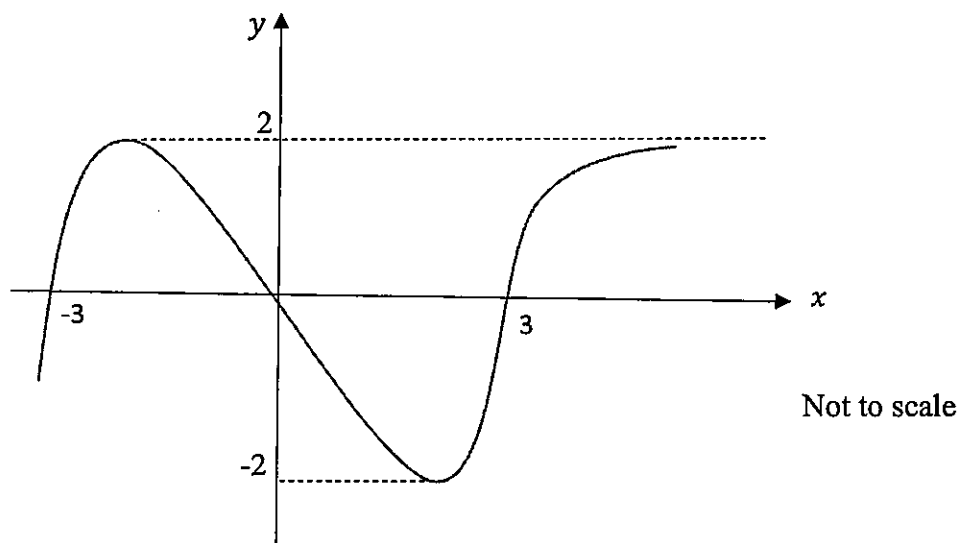
QUESTION 6**(Start a new sheet)****15 Marks**

a) i) Find constants a, b and c such that $\frac{x^2+2x}{(x^2+4)(x-2)} = \frac{ax+b}{x^2+4} + \frac{c}{x-2}$ 2

ii) Hence, find $\int \frac{x^2+2x}{(x^2+4)(x-2)} dx$ 2

b) Find $\int \frac{9x^3+9x^2+5x+4}{3x+1} dx$ 3

c) The diagram shows $y = f(x)$



Draw separate one third page sketch of the following graph

i) $y = f(x) + |f(x)|$ 2

ii) $y = \frac{1}{f(x)}$ 2

iii) $y^2 = f(x)$ 2

iv) $y = e^{f(x)}$ 2

QUESTION 7**(Start a new sheet)****15 Marks**

- a) i) Show that the equation of the tangent to the curve $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(x_1, y_1)$ on it is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$ 2

- ii) Hence, show that the equation of the chord of contact to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ from the point $R(x_0, y_0)$ is $\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$ 2

- iii) Hence, show that the chord of contact to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ from a point on the directrix is a focal chord 1

- b) $\int \sin(\ln x) dx$ using $u = \ln x$ or otherwise 3

- c) Evaluate $\int_0^{\ln 2} \frac{e^{2x}}{1+e^x} dx$ 3

- d) Let $z = \cos\theta + i\sin\theta$

- i) Show that $z^n + z^{-n} = 2\cos n\theta$, where n is a positive integer 2

- ii) Let m be a positive integer. Show that 2

$$(2\cos\theta)^{2m} = 2 \left[\cos 2m\theta + \binom{2m}{1} \cos(2m-2)\theta + \binom{2m}{2} \cos(2m-4)\theta + \dots + \binom{2m}{m-1} \cos 2\theta \right] + \binom{2m}{m}$$

QUESTION 8**(Start a new sheet)****15 Marks**

- a) i) By using the substitution $t = \tan \frac{x}{2}$, show that $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x} = 1$ 2
- ii) Show that $\int_0^a f(x) dx = \int_0^{\frac{a}{2}} \{f(x) + f(a - x)\} dx$ 2
- iii) Hence evaluate $\int_0^{\pi} \frac{x}{1 + \sin x} dx$ 1
- b) i) Find the real values of k for which the equation $\frac{x^2}{6-k} + \frac{y^2}{4-k} = 1$, defines respectively an ellipse and an hyperbola 2
- ii) Sketch the curve for $k=5$, showing its important features 2
- iii) Explain how the shape of this graph changes as k increases from 5 to 6 and determine if the graph has a limiting position for this increase. 1
- c) i) Show that the condition for the line $y = mx + c$ to be tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is that $c^2 = a^2 m^2 + b^2$ 2
- ii) Hence, or otherwise, prove that the pair of tangents from the point (3,4) to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ are at right angles to each other. 3

QUESTION 9**(Start a new sheet)****15 Marks**

a) Suppose k is a constant greater than 1. Let $f(x) = \frac{1}{1+(\tan x)^k}$, where $0 \leq x \leq \frac{\pi}{2}$.

You may assume $f\left(\frac{\pi}{2}\right) = 0$

i) Show that $f(x) + f\left(\frac{\pi}{2} - x\right) = 1$ for $0 \leq x \leq \frac{\pi}{2}$ 2

ii) Sketch $y = f(x)$ for $0 \leq x \leq \frac{\pi}{2}$ 3

** (There is no need to find $f'(x)$ but assume $y = f(x)$ has a horizontal tangent at $x = 0$.
Your graph should exhibit the property of a) i) and all of the information given)*

iii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1+(\tan x)^k}$ 2

b) For the curve $x^2y^2 - x^2 + y^2 = 0$

i) Demonstrate why $|y| < 1$ 1

ii) Demonstrate why $|y| \leq |x|$ 1

iii) Use implicit differentiation to show $\frac{dy}{dx} = \frac{x(1-y^2)}{y(1+x^2)}$ 2

iv) State the coordinates of any critical points (where the derivatives are undefined) 1

v) Explain why the curve has no stationary points 1

vi) Sketch the curve 2

QUESTION 10**(Start a new sheet)****15 Marks**

- a) Let $I_n = \int_0^{\frac{\pi}{2}} (\sin x)^n dx$ where n is an integer, $n \geq 0$.
- i) Using integration by parts, show that for $n \geq 2$, $I_n = \frac{(n-1)}{n} I_{n-2}$ 2
- ii) Evaluate I_4 2
- b) Suppose α, β, γ and δ are the four roots of the polynomial equation
 $P(x) = x^4 - Ax - 1 = 0$, where A is real
- i) By considering $P''(x)$, or otherwise, show that at most two of the roots are real 2
- ii) Prove that the polynomial equation with the four roots $\alpha^2, \beta^2, \gamma^2$ and δ^2 is $x^4 - 2x^2 - A^2x + 1 = 0$ 2
- iii) By considering another suitable polynomial equation, or otherwise, prove that $(\alpha^2 + 1)(\beta^2 + 1)(\gamma^2 + 1)(\delta^2 + 1) = A^2$ 2
- c) Let the points $A_1, A_2, \dots, A_n$ represent the n th roots of unity, $w_1, w_2, \dots, w_n$,
and suppose P represents any complex number z such that $|z| = 1$
- i) Show that $(PA_i)^2 = (z - w_i)(\bar{z} - \bar{w}_i)$ for $i = 1, 2, \dots, n$ 2
- ii) Prove that $\sum_{i=1}^n (PA_i)^2 = 2n$, given that $w_1 + w_2 + \dots + w_n = 0$ 3

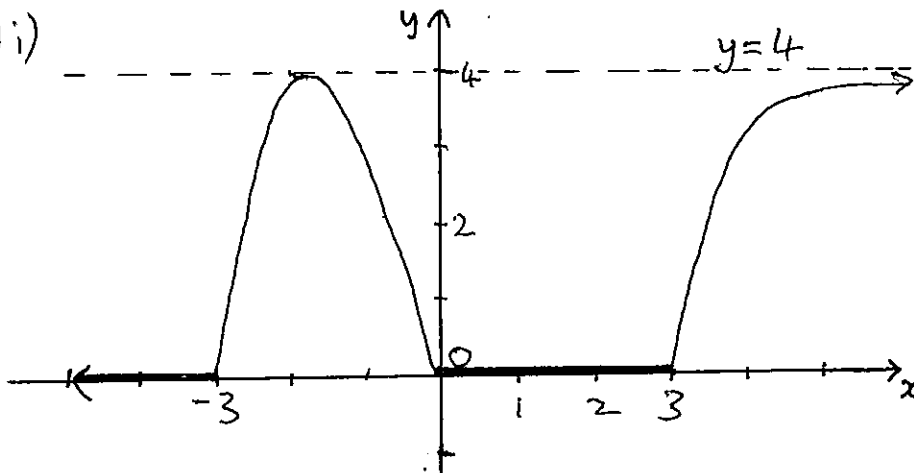
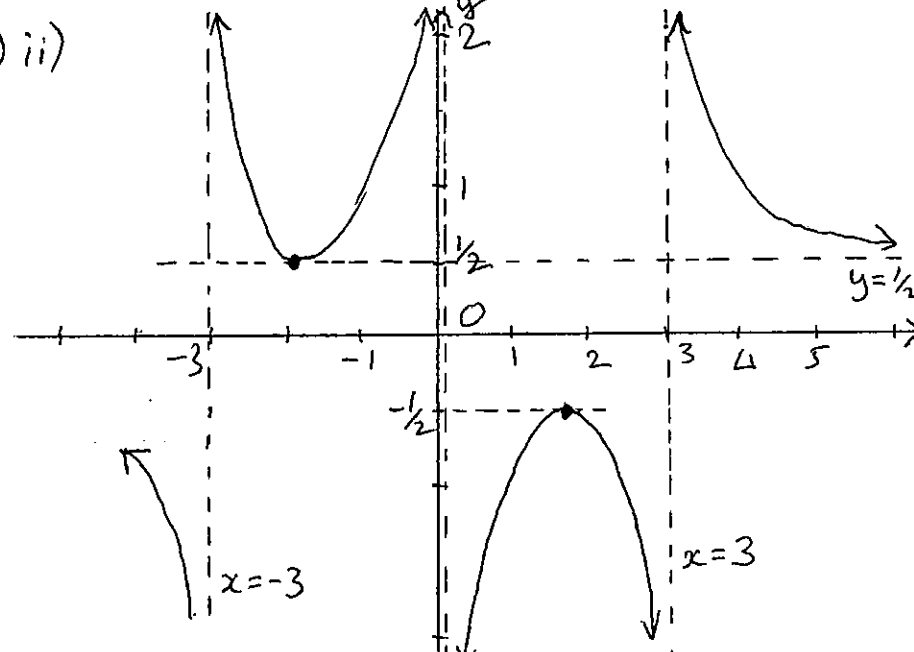
*******END OF EXAMINATION*******

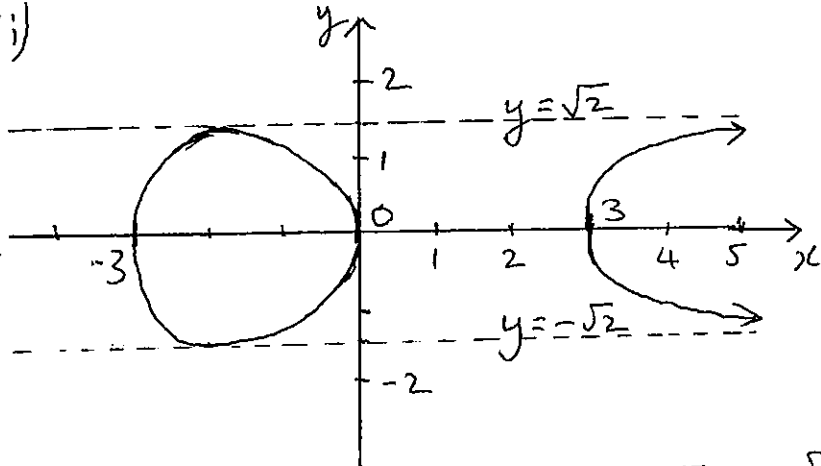
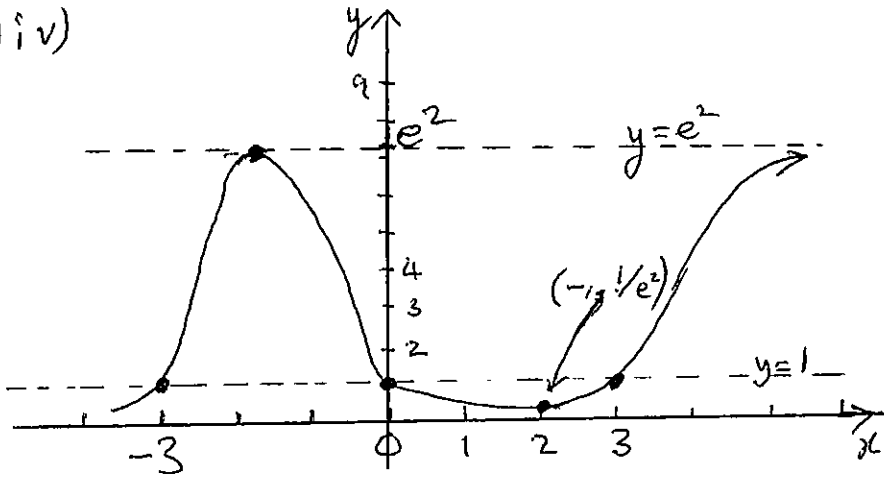
MULTIPLE CHOICE (1) B (2) D (3) B (4) D (5) C

MATHEMATICS Extension 2: Question...6...

Sheet 1 of 3

Suggested Solutions	Marks	Marker's Comments
6 a) i) $x^2 + 2x \equiv (x-2)(ax+b) + c(x^2+4)$ Substitute $x=2$: $8 = 8c \quad \therefore \underline{\underline{c=1}}$ Substitute $x=0$: $0 = -2b+4 \quad \therefore \underline{\underline{b=2}}$ Equate coefficients of x^2: $1 = a+1 \quad \therefore \underline{\underline{a=0}}$ $\therefore \underline{\underline{(a,b,c) = (0,2,1)}}$	2	One mark if only 2 values correct. (Essentially one mark off per error) Any combination of coeff. eqns and substitution would do.
ii) $\int \frac{x^2+2x dx}{(x^2+4)(x-2)} = \int \frac{2 dx}{x^2+4} + \int \frac{dx}{x-2}$ $= \underline{\underline{\tan^{-1}\left(\frac{x}{2}\right) + \ln x-2 + k}}$	2	One mark for each integral.
b) $ \begin{array}{r} 3x^2 + 2x + 1 \quad r \quad 3 \\ 3x+1 \overline{) 9x^3 + 9x^2 + 5x + 4} \\ \underline{9x^3 + 3x^2} \\ 6x^2 + 5x \\ \underline{6x^2 + 2x} \\ 3x + 4 \\ \underline{3x + 1} \\ 3 \end{array} $ (Long Division) $I = \int 3x^2 + 2x + 1 + \frac{3}{3x+1} dx$ $= \underline{\underline{x^3 + x^2 + x + \ln 3x+1 + k}}$	* 1 2	Note the use of absolute values in log integrals. Although marks were not deducted this time, Ext 2 students should use them when appropriate.

Suggested Solutions	Marks	Marker's Comments
c) i)  1. Maximum of $y=4$ 2. Labelled asymptote at $y=4$ 3. Cut of values at ± 3, 0 labelled 4. Some acknowledgement that $y=0$ for $x \leq -3$ and $0 \leq x < 3$.	2	Surprising number of people made no mark for $x \leq -3$ and $0 \leq x < 3$. The function exists there but equals 0. I need to see something different or some words.
c) ii)  1. Labelled asymptotes at $x=3$, $x=0$, $x=-3$, $y=1/2$ 2. Max and min values pinned to $\pm 1/2$ 3. Four parts of curve in correct alignment.	2	In this case (and (biv)) clearer with different scales. Equal scales preferable. Whichever is used, scales should be defined.

Suggested Solutions	Marks	Marker's Comments
c) iii)  1. Labelled asymptotes at $y = \sqrt{2}$, $y = -\sqrt{2}$ 2. Vertical lines (vertical tangents) at all x intercepts. 3. Points pinned as appropriate at $x = -3, 0, 3$ and $y = \pm 2$ 4. Symmetry in x axis.	2	Shape is always important. In this case it is important to show the verticals at the x intercepts, ($\frac{dy}{dx} \rightarrow \infty$)
c) iv)  1. Labelled asymptote at $y = e^2$ 2. Three crossings (marked some way) of the line $y = 1$ 3. Minimum value $1/e^2$ shown somehow 4. Maximum of e^2 shown somehow	2	little information for $x < -3$ but y certainly does not go negative. $e^2 \approx 7+$ It would be nice if that is shown in the scaling

Mathematics: Extension 1, Question 7

Suggested Solutions	Marks	Marker's Comments
(a) (i) To find equation of tangent to curve at $P(x_1, y_1)$: $\frac{d}{dx} \left(\frac{x^2}{a^2} - \frac{y^2}{b^2} \right) = \frac{d}{dx} (1)$ $\frac{2x}{a^2} - \frac{2yy'}{b^2} = 0$ $y' = \frac{b^2 x}{a^2 y}$ At $P(x_1, y_1)$, the gradient m is therefore, $m = \frac{b^2 x_1}{a^2 y_1}$ Hence the equation of the tangent to the hyperbola at P is $y - y_1 = \frac{b^2 x_1}{a^2 y_1} (x - x_1)$ $a^2 y_1 y - a^2 y_1^2 = b^2 x_1 x - b^2 x_1^2$ $b^2 x_1^2 - a^2 y_1^2 = b^2 x x_1 - a^2 y y_1$ $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = \frac{x x_1}{a^2} - \frac{y y_1}{b^2} \quad \text{after dividing through by } a^2 b^2$ Since $P(x_1, y_1)$ lies on the hyperbola, $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$, and so $\frac{x x_1}{a^2} - \frac{y y_1}{b^2} = 1$ as required. (ii) Let the tangents through $R(x_0, y_0)$ meet the curve at points $P(x_1, y_1)$ and $Q(x_2, y_2)$. The tangent at P and Q are given by the result of (i) respectively by $\frac{x x_1}{a^2} - \frac{y y_1}{b^2} = 1 \quad \dots (1)$ and $\frac{x x_2}{a^2} - \frac{y y_2}{b^2} = 1 \quad \dots (2)$	1 for gradient. 1 for tangent. 1 for acquiring tangent at Q and explaining correctly that (3) and (4) follow because R lies on both lines.	Should state why we obtain '1' (i.e. P satisfies the equation). Tangent at Q needn't be derived, but at least quote 'similarly' for Q as P.

Since each tangent passes through R , we must have by (1) and (2),

$$\frac{x_0x_1}{a^2} - \frac{y_0y_1}{b^2} = 1 \quad (3)$$

$$\frac{x_0x_2}{a^2} - \frac{y_0y_2}{b^2} = 1 \quad (4)$$

Now, consider the line l :

$$l: \frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$$

Then by (3), P lies on l , and by (4), Q lies on l .

Since two points determine a line uniquely, the chord PQ lies on l . Hence the chord (of contact) is given by

$$\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$$

- (iii) Let $(\pm \frac{a}{e}, y_0)$ be an arbitrary point on one of the directrices. The chord of contact from this point to the hyperbola is then

$$\frac{x(\pm \frac{a}{e})}{a^2} - \frac{yy_0}{b^2} = 1$$

$$\pm \frac{x}{ae} - \frac{yy_0}{b^2} = 1 \quad (5)$$

The chord passes through the corresponding focus if and only if $(\pm ae, 0)$ satisfies equation (5):

$$\begin{aligned} LHS_{of(5)} &= \pm \frac{\pm ae}{ae} - \frac{y(0)}{b^2} \\ &= 1 - 0 \\ &= 1 \\ &= RHS_{of(5)} \end{aligned}$$

Hence, the chord of contact from a point on the directrix is a focal chord (passing through the corresponding focus).

1 for reasoning correctly why the equation required was that containing the chord of contact.

1 for showing that the chord passing through arbitrary point on directrix was focal chord.

Did not penalise for assuming $(\frac{a}{e}, y_0)$ only.

Many students constructed the two equations (3) and (4) correctly, but then failed to state (logically) how this could lead to the chord of contact equation.

The first two equations are results that we keep to the side. We then consider a line of form given by line l . THEN, since equations (3) and (4) SHOW that P, Q satisfy l , and since there is only one line passing through two points, it MUST be the case that l is the equation of a line connecting P and Q ; i.e. a line containing the chord of contact (a line segment).

Many students assumed only a point on the directrix of the positive arm of the hyperbola. If this were a 2-mark question, one mark would not have been awarded.

Also, too many students did not select an arbitrary point on the directrix (as the external point for chord of contact), and then use that with chord of contact equation to derive the appropriate equation.

Once you have the chord of contact for

(b) In

$$\int \sin(\log x) dx$$

let $u = \log x$. Then $x = e^u$ and so $dx = e^u du$, giving

$$\int \sin(u) e^u du$$

Now, let $U = \sin u$ and $dV = e^u du$. Then $dU = \cos u du$ and $V = e^u$, so that:

$$\int \sin(u) e^u du = e^u \sin u - \int \cos(u) e^u du$$

For

$$\int \cos(u) e^u du$$

let $U = \cos u$ and $dV = e^u du$. Then $dU = -\sin u du$ and $V = e^u$, so,

$$\begin{aligned} \int \sin(u) e^u du &= e^u \sin u - \int \cos(u) e^u du \\ &= e^u \sin u - \left[e^u \cos u - \int (-\sin u) e^u du \right] \\ &= e^u \sin u - e^u \cos u - \int \sin(u) e^u du \end{aligned}$$

So,

$$2 \int \sin(u) e^u du = e^u \sin u - e^u \cos u$$

giving

$$\int \sin(\log x) dx = \frac{x}{2} (\sin(\log x) - \cos(\log x)) + C$$

this situation, you're then tasked with proving that it passes through the focal point. That is, you're being asked to show that **every** such chord passes through the appropriate focus. Choosing $(\frac{a}{e}, 0)$ as your external point does not do this.

The integration was handled well.

1 for correct integrand after substitution.

The only cause for concern was that of students not keeping track of their working/brackets, and letting trouble with minus signs ensue.

1 for a correct intermediate integral.

Also, you must give your final answer in terms of the original variable – don't leave you integral in substituted form.

Also, do not forget to include a constant of integration. No penalisation here for such instances, but, there are situations in a mathematics problem where the constant of integration is important (e.g. you may have to derive an original function) – leaving it off leads to errors.

1 for correct final result (must have substituted back for original variable).

(c)

$$\int_0^{\log 2} \frac{e^{2x}}{1+e^x} dx$$

Let $u = 1 + e^x$. Then $u = 2$ when $x = 0$ and $u = 3$ when $x = \log 2$.

Also,

$$du = e^x dx \Rightarrow \frac{du}{e^x} = dx \Rightarrow \frac{du}{u-1} = dx. \text{ So,}$$

$$\int_0^{\log 2} \frac{e^{2x}}{1+e^x} dx = \int_2^3 \frac{(u-1)^2}{u} \frac{du}{(u-1)}$$

$$= \int_2^3 \frac{u-1}{u} du$$

$$= \int_2^3 1 - \frac{1}{u} du$$

$$= u - \log u \Big|_2^3$$

$$= (3 - \log 3) - (2 - \log 2)$$

$$= 1 + \log 2 - \log 3$$

$$= 1 + \log \frac{2}{3}$$

'or otherwise...'

$$\int_0^{\log 2} \frac{e^{2x}}{1+e^x} dx = \int_0^{\log 2} \frac{e^x}{1+e^x} de^x$$

$$= \int_0^{\log 2} \frac{1+e^x-1}{1+e^x} de^x$$

$$= \int_0^{\log 2} 1 - \frac{1}{1+e^x} de^x$$

$$= e^x - \log(1+e^x) \Big|_0^{\log 2}$$

$$= (2 - \log 3) - (1 - \log 2)$$

$$= 1 + \log \frac{2}{3}$$

The integration was handled well.

Errors crept in where people did not take care with the algebra once substitutions were made.

There were also errors with signs (again), leading to incorrect final answer.

1 for a correct integral.

1 for correct integration.

1 for correct result.

A sizeable portion of the candidature approached the problem using techniques illustrated left.

It is mathematically 'correct', but should **not be used for the HSC**.

In this integral, we choose to integrate with respect to the function e^x . Note that $de^x = e^x dx$, so we recover the original integrand.

(d)

(i) Let $z = \cos \theta + i \sin \theta$.

Then for $n \in \mathbb{N}_0$,

$$\begin{aligned} z^n + z^{-n} &= (\cos \theta + i \sin \theta)^n + (\cos \theta + i \sin \theta)^{-n} \\ &= (\cos n\theta + i \sin n\theta) + (\cos(-n\theta) + i \sin(-n\theta)) \quad \dots (*) \\ &= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ &= 2 \cos n\theta \end{aligned}$$

where (*) follows by De Moivre's Theorem.

(ii) $(2 \cos \theta)^{2m} = (z + z^{-1})^{2m}$ by (i) with $n = 1$.

Now, by the binomial theorem,

$$\begin{aligned} (2 \cos \theta)^{2m} &= (z + z^{-1})^{2m} = \sum_{k=0}^{2m} \binom{2m}{k} z^{2m-k} (z^{-1})^k \\ &= \sum_{k=0}^{2m} \binom{2m}{k} z^{2m-2k} \quad \dots (1) \end{aligned}$$

Note that there are an **odd** number of terms (counting begins at zero and ends on an even number). When $k = m$, we have

$$\binom{2m}{m} z^{2m-2m} = \binom{2m}{m} z^0 = \binom{2m}{m}$$

For the remaining $2m$ other terms, we can pair the k^{th} term with its symmetrical partner, the $(2m - k)^{\text{th}}$ term, as follows:

$$\begin{array}{l|l} \binom{2m}{k} z^{2m-2k} & \binom{2m}{2m-k} z^{2m-2(2m-k)} \\ & = \binom{2m}{2m-k} z^{-(2m-2k)} \\ & = \binom{2m}{k} z^{-(2m-2k)} \quad (\text{using } \binom{n}{r}) \\ & = \binom{n}{n-r} \end{array}$$

As there are $2m$ terms being paired, we have m pairings, and hence a sum from $k = 0$ to $m - 1$.

1 for correct use of De Moivre's Theorem.

1 for correct result.

1 for correct use of binomial theorem, leading to something of consequence to the solution of the problem.

1 for remainder of complete solution.

No problems with using De Moivre's theorem to prove the result in (i).

Many of those who attempted part (ii) recognised the symmetry involved and demonstrated so.

It was important to show why $\binom{2m}{m}$ appeared as a lone term, and explain the consequence of symmetry here (Pascal's relation).

Some students provided working, but not enough in the sense that, to obtain the final mark, you had to show directly how the result followed from your working. This is important because you're given the solution to the problem (if you weren't given the solution, and you produced the correct answer, the assessor would give you the mark. Without steps, you're asking the marker to essentially 'believe you' when you make the leap

So,

$$\begin{aligned}(2 \cos \theta)^{2m} &= (z + z^{-1})^{2m} = \sum_{k=0}^{2m} \binom{2m}{k} z^{2m-2k} \\&= \sum_{k=0}^{m-1} \binom{2m}{k} (z^{2m-2k} + z^{-(2m-2k)}) + \binom{2m}{m} \\&= \sum_{k=0}^{m-1} \binom{2m}{k} (2 \cos(2m - 2k)\theta) + \binom{2m}{m} \quad \text{by result of part (i)} \\&= 2 \sum_{k=0}^{m-1} \binom{2m}{k} \cos(2m - 2k)\theta + \binom{2m}{m}\end{aligned}$$

as required.

from limited working
to the desired
conclusion).

MATHEMATICS Extension 2 : Question..8...

Suggested Solutions	Marks	Marker's Comments
(a) $t = \tan \frac{x}{2} \quad \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$ $\frac{dx}{dt} = \frac{2}{\sec^2 \frac{x}{2}}$ $dx = \frac{2 dt}{1+t^2}$ $x = 0 \quad t = 0$ $x = \pi/2 \quad t = 1$ $0 \sim \pi/2$ $0 \sim 1$ Confusing. $\int_0^1 \frac{2 dt}{1+t^2} dt$ $= \int_0^1 \frac{2 dt}{(1+t)^2}$ $= 2 \left[-\frac{1}{1+t} \right]_0^1$ $= 2 \left[-\frac{1}{2} + 1 \right]$ $= 1$	1	t formula on reference sheet no excuse for getting incorrect. Be careful of notation
(ii) Method 1 $\int_0^a f(x) dx = \int_0^{a/2} f(x) dx + \int_{a/2}^a f(x) dx$ let $x = a - u \quad dx = -du$ $x = \frac{a}{2} \quad u = \frac{a}{2}$ $x = a \quad u = 0$ $= \int_0^{a/2} f(x) dx - \int_{a/2}^0 f(a-u) du$ $= \int_0^{a/2} f(x) dx + \int_0^{a/2} f(a-x) dx$ with dummy variable	1	Needed to see $\int_0^a f(a-u) du$ step $a/2$ for second mark.

MATHEMATICS Extension 2 : Question 8

Suggested Solutions

Marks

Marker's Comments

Method 2

Let primitive of $f(x)$ be $F(x)$

$$\int_0^a f(x) dx = F(a) - F(0)$$

RHS =

$$\int_0^{a/2} f(x) dx + \int_0^{a/2} f(a-x) dx$$

$$= F(a/2) - F(0) + \left[F(a - a/2) - F(a) \right]$$

$$= F(a) - F(0)$$

$$= \text{LHS}$$

Method 3 was method 1 backwards.

Students who used property $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

used it incorrectly.

(c) $\int_0^{\pi} \frac{x}{1+\sin x} dx$ from part (c)

$$= \int_0^{\pi/2} \frac{x}{1+\sin x} + \frac{\pi-x}{1+\sin(\pi-x)} dx$$

$$= \int_0^{\pi/2} \frac{x+\pi-x}{1+\sin x} dx \quad \text{since } \sin(\pi-x) = \sin x$$

$$= \int_0^{\pi/2} \frac{\pi}{1+\sin x} dx \quad \text{from part (c)}$$

$$= \pi$$

Students didn't look closely and used $\pi/2 - x$.

MATHEMATICS Extension 2: Question...8...

Suggested Solutions

Marks

Marker's Comments

$$b(i) \frac{x^2}{6-k} + \frac{y^2}{4-k} = 1$$

to be ellipse must be of form.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

$$\therefore 6-k > 0 \text{ and } 4-k > 0$$

$$k < 6 \text{ and } k < 4$$

$$\text{so } \underline{k < 4}$$

For hyperbola

$$6-k > 0 \text{ and } 4-k < 0$$

$$k < 6 \text{ and } k > 4.$$

$$4 < k < 6$$

OR

$$\left(\begin{array}{l} 6-k < 0 \text{ and } 4-k > 0 \\ k \geq 6 \text{ and } k < 4 \\ \text{so no solution} \end{array} \right)$$

(ii) see over.

(iii) As k increases from 5 to 6 graph becomes steeper vertex approaches origin and hyperbola approaches the vertical line $x=0$ or y -axis

Students who looked at the middle question much harder and didn't get correct answer

didn't lose marks for not considering this.

Needed to indicate it was becoming vertical and approaching 0. partly done.

MATHEMATICS Extension 2: Question...5....

Suggested Solutions

Marks

Marker's Comments

$$(ai) \quad a^2 = b^2(e^2 - 1)$$

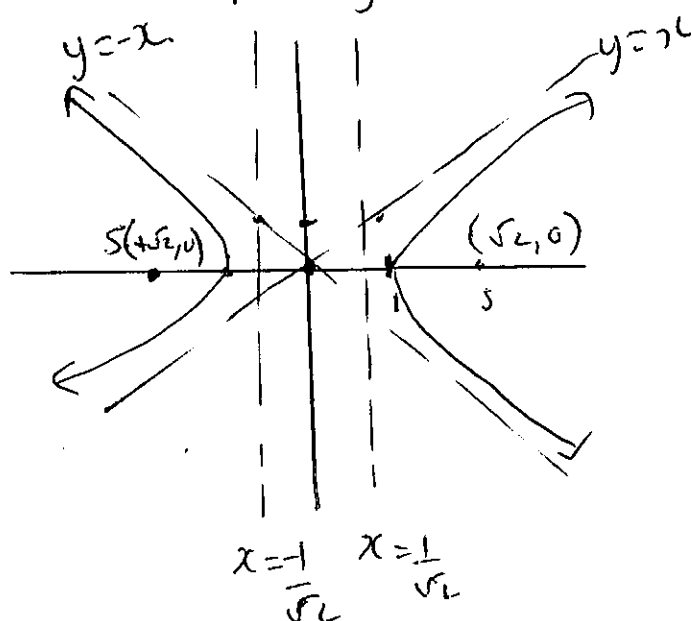
$$1 = 1(e^2 - 1)$$

$$e^2 = 2$$

$$e = \sqrt{2}$$

$$b: x = \pm \frac{1}{\sqrt{2}} \quad \text{focus } (\pm\sqrt{2}, 0)$$

$$\text{asymptote } y = \pm x.$$



1 for hyperbola
and asymptotes

1 for foci,
and directrices
and $e = \sqrt{2}$.

MATHEMATICS Extension 2 : Question 8.....

Suggested Solutions	Marks	Marker's Comments
c) (i) $\frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$ $\therefore b^2 x^2 + a^2 (m^2 x^2 + 2mcx + c^2) = a^2 b^2$ $\therefore (b^2 + a^2 m^2) x^2 + 2ma^2 c x + a^2 c^2 - a^2 b^2 = 0$ for tangent $\Delta = 0$ $4a^4 m^2 c^2 - 4(b^2 + a^2 m^2)(a^2 c^2 - a^2 b^2) = 0$ $a^2 m^2 c^2 - b^2 c^2 + b^4 - a^2 m^2 c^2 + a^2 m^2 b^2 = 0$ $\therefore \cancel{a^2 m^2 c^2} - b^2 c^2 + b^4 - \cancel{a^2 m^2 c^2} + a^2 m^2 b^2 = 0$ $\therefore -b^2 c^2 + b^4 + a^2 m^2 b^2 = 0$ $c^2 = b^2 + a^2 m^2$ as required (ii) $\frac{x^2}{16} + \frac{y^2}{9} = 1 \quad c^2 = 16m^2 + 9$ at (3, 4) $4 = 3m + c$ $\therefore c = 4 - 3m$ $(4 - 3m)^2 = 16m^2 + 9$ $7m^2 - 24m - 7 = 0$. Roots are m_1 and m_2 products of roots $= -\frac{7}{7}$ $= -1$ $\therefore m_1 m_2 = -1$ tangents are perpendicular	1 1 1	well done some poor algebra. some people went off on wrong path. by finding equation of tangent. 8. Some solved quadratic then multiplied and

Suggested Solutions

$$\begin{aligned} \text{a) ; } f\left(\frac{\pi}{2} - x\right) &= \frac{1}{1 + \tan\left(\frac{\pi}{2} - x\right)^k} \\ &= \frac{1}{1 + \frac{\sin^k\left(\frac{\pi}{2} - x\right)}{\cos^k\left(\frac{\pi}{2} - x\right)}} \\ &= \frac{\cos^k\left(\frac{\pi}{2} - x\right)}{\cos^k\left(\frac{\pi}{2} - x\right) + \sin^k\left(\frac{\pi}{2} - x\right)} \\ &= \frac{\sin^k x}{\sin^k x + \cos^k x} \end{aligned}$$

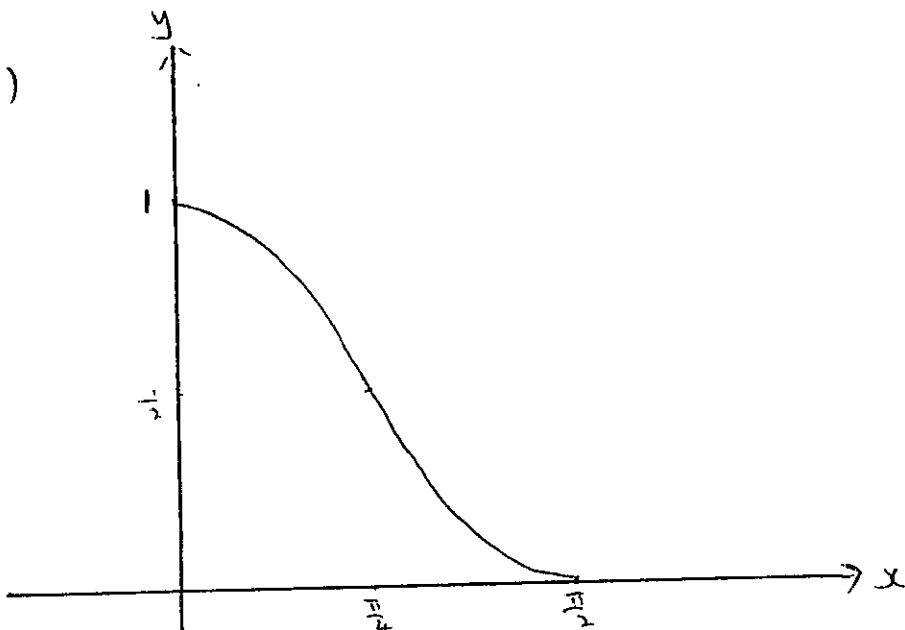
(1)

$$\begin{aligned} f(x) &= \frac{1}{1 + \frac{\sin^k x}{\cos^k x}} \\ &= \frac{\cos^k x}{\cos^k x + \sin^k x} \end{aligned}$$

$$\begin{aligned} \therefore f(x) + f\left(\frac{\pi}{2} - x\right) &= \frac{\cos^k x + \sin^k x}{\cos^k x + \sin^k x} \\ &= 1 \end{aligned}$$

(1)

ii)



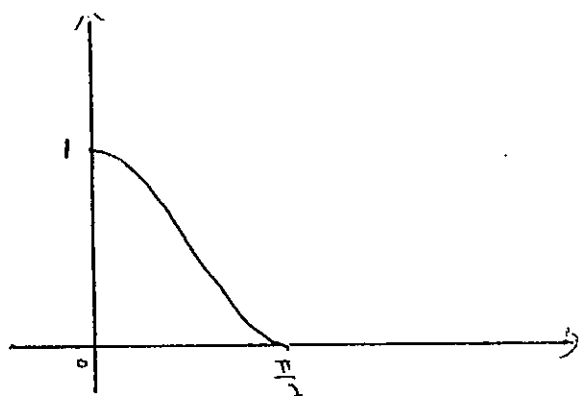
3 marks

Suggested Solutions

Marks

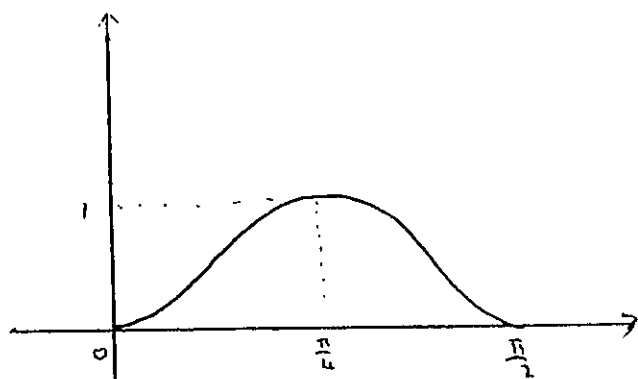
Marker's Comments

Other graphs.



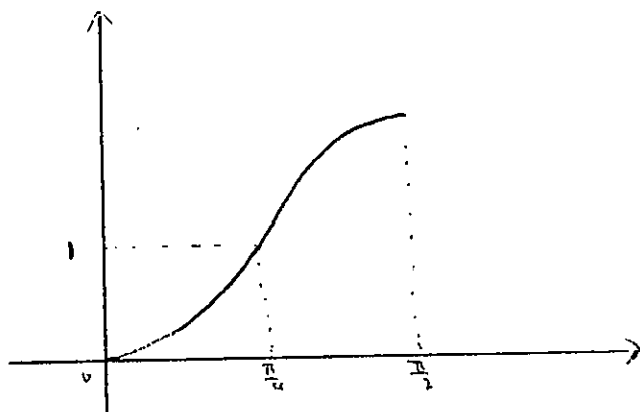
2 marks

no $(\frac{\pi}{4}, \frac{1}{2})$



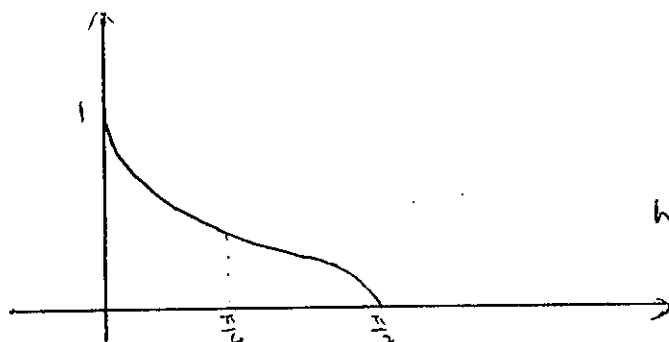
1 mark

(wrong y-intercept)



2 marks

(back to front)

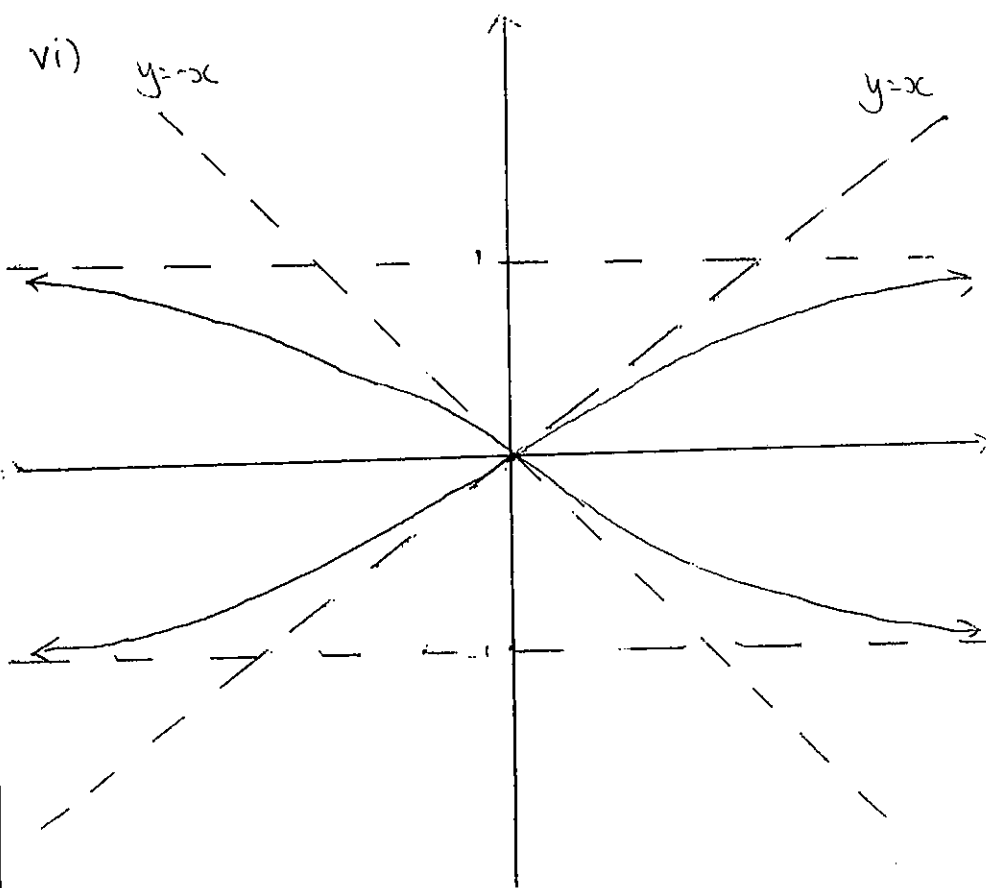


2 marks

wrong concavity

Suggested Solutions	Marks	Marker's Comments
iii) <u>Method 1</u> Since Area is exactly half the area of rectangle with length $\frac{\pi}{2}$ and height 1. (By graph) $\therefore I = \frac{1}{2} \times \frac{\pi}{2} \times 1$ $= \frac{\pi}{4}$ iv) Let $I = \int_0^{\frac{\pi}{2}} \frac{1}{1 + \tan^k x} dx$ $= \int_0^{\frac{\pi}{2}} \frac{1}{1 + \tan^k(\frac{\pi}{2} - x)} dx$ $\therefore 2I = \int_0^{\frac{\pi}{2}} 1 dx$ (from i) ————— $\therefore 2I = [x]_0^{\frac{\pi}{2}}$ $I = \frac{\pi}{2} \times \frac{1}{2}$ $= \frac{\pi}{4}$ Anyone who wrote $2I = 1$ and $\therefore I = \frac{1}{2}$ received 0 marks!	(1) (1)	

Suggested Solutions	Marks	Marker's Comments
b) i) $x^2 y^2 - x^2 + y^2 = 0$ $x^2 y^2 + y^2 = x^2$ $y^2(x^2 + 1) = x^2$ $y^2 = \frac{x^2}{x^2 + 1}$ < 1 (since $x^2 + 1 > x^2$) $\therefore y < 1$		①
ii) $x^2 y^2 - x^2 + y^2 = 0$ $\therefore x^2 - y^2 = x^2 y^2$ ≥ 0 $\therefore x^2 - y^2 \geq 0$ $x^2 \geq y^2$ $y^2 \leq x^2$ $ y \leq x $	Many assumed that $\frac{y^2}{1-y^2} \geq 1$ and $\therefore x \geq 1$ <u>Wrong!!!</u>	①
iii) $2xy^2 + 2y \frac{dy}{dx} x^2 - 2x + 2y \frac{2y}{dx} = 0$ $\frac{dy}{dx} (2yx^2 + 2y) - (2x - 2xy^2) = 0$ $\frac{dy}{dx} (2yx^2 + 2y) = 2x - 2xy^2$ $\frac{dy}{dx} = \frac{2x - 2xy^2}{2yx^2 + 2y}$ $= \frac{x(1-y^2)}{y(1+x^2)}$		①

Suggested Solutions	Marks	Marker's Comments
iv) $\frac{dy}{dx}$ is undefined with $y(1+x^2) = 0$ $\therefore x = 0$ When $y = 0$, $x = 0$ $\therefore$ critical point $(0, 0)$ v) $\frac{dy}{dx} = 0 \rightarrow x(1-y^2) = 0$ $\therefore x = 0$ or $y = \pm 1$ Since $\frac{dy}{dx}$ at $(0, 0)$ is undefined (part iv) and $y < 1$, $\therefore y \neq \pm 1$. There are no stationary points. vi) 		(1) 1 - shape 1 - lying between $y = \pm 1$ and $y = \pm x$.

MATHEMATICS Extension 2: Question 10

Suggested Solutions	Marks	Marker's Comments
a). (i) $I_n = \int_0^{\pi/2} \sin^n x dx = \int_0^{\pi/2} \sin^{n-1} x \sin x dx$ Let $u = \sin^{n-1} x \Rightarrow du = (n-1) \sin^{n-2} x \cos x dx$ and $dv = \sin x dx \Rightarrow v = -\cos x$		<u>Method I</u>
$\therefore I_n = \sin^{n-1} x (-\cos x) \Big _0^{\pi/2} - (n-1) \int_0^{\pi/2} (-\cos x) \sin^{n-2} x \cos x dx \dots$ (by parts) $= 0 + (n-1) \int_0^{\pi/2} \cos^2 x \sin^{n-2} x dx$ $= (n-1) \int_0^{\pi/2} (1 - \sin^2 x) \sin^{n-2} x dx$ $= (n-1) \int_0^{\pi/2} (\sin^{n-2} x - \sin^n x) dx$ $= (n-1) \left[\int_0^{\pi/2} \sin^{n-2} x dx - \int_0^{\pi/2} \sin^n x dx \right]$ $= (n-1) [I_{n-2} - I_n]$ $I_n + (n-1) I_n = (n-1) I_{n-2}$ $n I_n = (n-1) I_{n-2}$ $\therefore I_n = \frac{(n-1)}{n} I_{n-2}$ as required	1 1	Must show what happens to $-\sin^{n-1} x \cos x \Big _0^{\pi/2}$ to gain maximum marks For recognising that $I_n = \int_0^{\pi/2} \sin^n x dx$ and $I_{n-2} = \int_0^{\pi/2} \sin^{n-2} x dx$

TRAHS : Maths Extension 2 : Task 2 Term 1 2018

MATHEMATICS Extension 2: Question 10

Suggested Solutions

Marks

Marker's Comments

Method 2 :

$$I_n = \int_0^{\pi/2} \sin^{n-1} x \sin x dx$$

$$= \int_0^{\pi/2} \sin^{n-1} x \cdot \frac{d}{dx} (-\cos x) dx$$

$$= \sin^{n-1} x (-\cos x) \Big|_0^{\pi/2} - \int_0^{\pi/2} -\cos x \frac{d}{dx} (\sin^{n-1} x) dx \dots (\text{by parts})$$

$$= 0 + (n-1) \int_0^{\pi/2} \cos^2 x \cdot \sin^{n-2} x dx \quad 1$$

$$= (n-1) \int_0^{\pi/2} (1 - \sin^2 x) \sin^{n-2} x dx$$

$$= (n-1) \int_0^{\pi/2} (\sin^{n-2} x - \sin^n x) dx$$

$$= (n-1) [I_{n-2} - I_n] \quad 1$$

$$\therefore I_n + (n-1)I_n = (n-1)I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n} I_{n-2}$$

TRAHS : Maths Ext 2 : Task 2 Term 1 2018

MATHEMATICS Extension 2: Question 10

Suggested Solutions

Marks

Marker's Comments

$$a_{(ii)} \quad I_n = \int_0^{\pi/2} \sin^n x dx \quad \dots \text{given}$$

$$I_n = \frac{n-1}{n} I_{n-2} \quad \dots \text{proved above}$$

$$\therefore I_4 = \frac{3}{4} I_2$$

$$= \frac{3}{4} \left(\frac{1}{2} I_0 \right)$$

$$= \frac{3}{8} \int_0^{\pi/2} \sin^0 x dx$$

$$= \frac{3}{8} \int_0^{\pi/2} 1 \cdot dx$$

$$= \frac{3}{8} (x) \Big|_0^{\pi/2}$$

$$= \frac{3\pi}{16}$$

Note: $I_2 = \int_0^{\pi/2} \sin^2 x dx$ could have been used to get $\pi/4$

1

First mark for getting

$$I_2 = \frac{\pi}{4}$$

1

2nd mark for getting

$$I_4 = \frac{3\pi}{16}$$

MATHEMATICS Extension 2: Question 10

Suggested Solutions

Marks

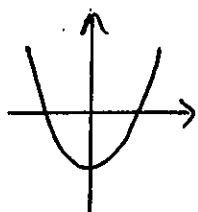
Marker's Comments

b) Method I :

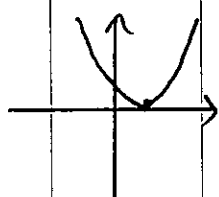
(i) $P(x) = x^4 - Ax - 1$

$P'(x) = 4x^3 - A$

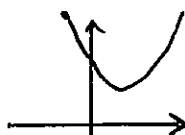
$P''(x) = 12x^2$

 $\Rightarrow P''(x) > 0$ for all $x \in \mathbb{R}$ $\Rightarrow$ curve is concave up

distinct roots



equal roots

real
no roots

1

For concluding
curve is concave
up for all x

1

For demonstrating
that there could
be 2 distinct
roots, a double
root, or no
real roots $\therefore$ At most there are 2 real rootsFor 2nd mark, there is only
1 turning point at $x = \sqrt[3]{A/4}$ OR

$P''(x) = 0$, when $x = 0$

But $P(0) \neq 0$, hence no
double roots and with 2 complex
roots (conjugate pairs as coefficients are real), theremust be at most
2 real roots

MATHEMATICS Extension 2: Question 10

Suggested Solutions

Marks

Marker's Comments

b) Method III:

(i)

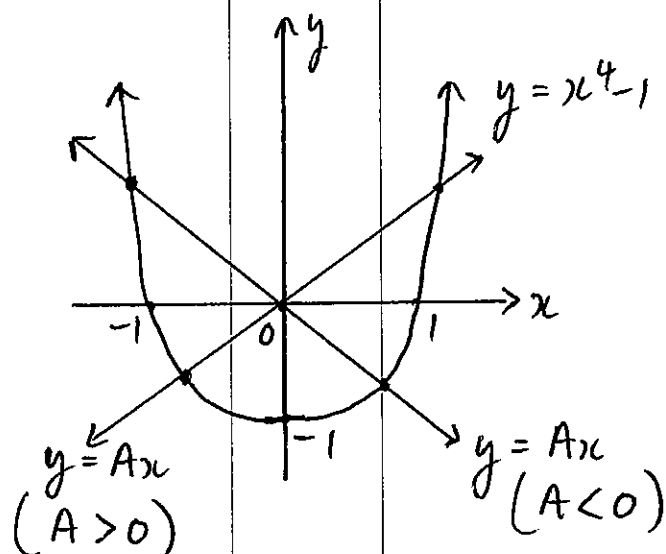
$$P(x) = x^4 - Ax - 1$$

$$P(x) = 0$$

$\Rightarrow$

$$x^4 - 1 = Ax$$

Graph $y = x^4 - 1$ and $y = Ax$



1

1

clearly there are 2 points of intersection when $x^4 - 1 = Ax$

and hence $P(x) = x^4 - Ax - 1$

has at most 2 real roots

MATHEMATICS Extension 2: Question 10

Suggested Solutions

Marks

Marker's Comments

b) Method II:

(1) $P(x) = x^4 - Ax - 1, A \in \mathbb{R}$

Clearly polynomial with real coefficients
 $\Rightarrow P(x)$ can be expressed as a product
 of linear and quadratic irreducible
 factors, by ^{OR} FTA

1st mark

But $P(x)$ = quartic polynomial
 (at most 4 roots)

• 4 complex roots : (quadratic irreducible)²

• 2 complex
 +
 double root : (linear)² x quadratic
 irreducible

• 2 complex
 +
 2 distinct
 roots : linear x linear x quadratic
 irreducible

2nd mark

clearly this quartic polynomial
 has at most 2 real roots

JRAHS: Ext 2 Task 2 Term 1 2018

MATHEMATICS Extension 2: Question 10

Suggested Solutions	Marks	Marker's Comments
b ii) If $\alpha, \beta, \gamma, \delta$ are the roots of $x^4 - Ax - 1 = 0$, then $\alpha^2, \beta^2, \gamma^2, \delta^2$ must satisfy $(x^{\frac{1}{2}})^4 - A(x^{\frac{1}{2}}) - 1 = 0$ $\Rightarrow x^2 - A\sqrt{x} - 1 = 0$ $x^2 - 1 = A\sqrt{x}$ $(x^2 - 1)^2 = A^2 x$ $\therefore x^4 - 2x^2 - A^2 x + 1 = 0 \text{ is the required equation}$	1 1	For making Transition to the new equation For algebraic manipulation Note: if $m = \alpha^2$ then $\alpha = \pm\sqrt{m}$
(iii) Since $\alpha^2, \beta^2, \gamma^2, \delta^2$ are roots of $x^4 - 2x^2 - A^2 x + 1 = 0$, then $\alpha^2 + 1, \beta^2 + 1, \gamma^2 + 1, \delta^2 + 1$ must satisfy $(x-1)^4 - 2(x-1)^2 - A^2(x-1) + 1 = 0$ But product of roots $= (\alpha^2+1)(\beta^2+1)(\gamma^2+1)(\delta^2+1) = \frac{e}{a}$ $= (-1)^4 - 2(-1)^2 + A^2 + 1$ $= 1 - 2 + A^2 + 1$ $= A^2 \text{ as required}$	1 1	For transition to new equation Algebraic manipulation

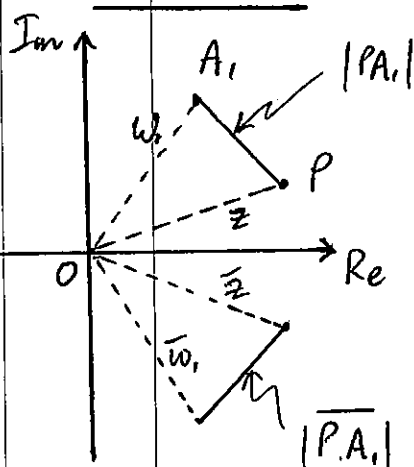
Note: If $x = \pm\sqrt{m-1}$, then $x^4 - Ax - 1 = 0$ is the equation into which $\pm\sqrt{m-1}$ needs to be substituted

QUESTION 10

(c)

Note:

$$\begin{aligned} PA_i &= PO + OA_i \\ &= -\bar{z} + w_i \\ &= w_i - \bar{z} \end{aligned}$$



$$(i) |PA_i| = |z - w_i|$$

$$\therefore |PA_i|^2 = |z - w_i|^2$$

$$\text{or } PA_i^2 = |z - w_i|^2$$

$$= (z - w_i)(\bar{z} - \bar{w}_i) \text{ since } |z|^2 = z \cdot \bar{z}$$

$$= (z - w_i)(\bar{z} - \bar{w}_i)$$

for $i = 1, 2, \dots, n$

$$\text{If RHS} = (z - w_i)(\bar{z} - \bar{w}_i) \quad \text{must say why!}$$

$$= (z - w_i)(\bar{z} - \bar{w}_i)$$

$$= |z - w_i|^2$$

must say why!

$$= (z - w_i)^2$$

$$= (PA_i)^2$$

must say why!

to get maximum marks.

for linking the distance PA_i to $|z - w_i|$ and using $|z|^2 = z \cdot \bar{z}$

and using mod of the difference = difference of the mods.

Note : must mention

$$|z|^2 = z \cdot \bar{z}$$

and

$$\overline{(z - w_i)} = (\bar{z} - \bar{w}_i)$$

to get maximum marks.

Question 10

$$(c) (ii) \sum_{i=1}^n |A_i|^2 = \sum_{i=1}^n (z - w_i)(\bar{z} - \bar{w}_i) \dots$$

from (i)

$$= \sum_{i=1}^n (z\bar{z} - z\bar{w}_i - \bar{z}w_i + w_i\bar{w}_i)$$

$$= \sum_{i=1}^n (1 - z\bar{w}_i - \bar{z}w_i + 1) \dots$$

$$= \sum_{i=1}^n (2 - z\bar{w}_i - \bar{z}w_i)$$

$$= 2n - z \sum_{i=1}^n \bar{w}_i - \bar{z} \sum_{i=1}^n w_i$$

$$= 2n - z(\bar{w}_1 + \bar{w}_2 + \dots + \bar{w}_n) - \bar{z}(w_1 + w_2 + \dots + w_n)$$

$$= 2n - z(0) - \bar{z}(0)$$

$$= 2n - 0 - 0$$

$$= 2n$$

as required

For expanding

AND

For using

$$z\bar{z} = |z|^2 = 1$$

$$w_i\bar{w}_i = |w_i|^2 = 1$$

For recognising that $z\bar{z}$ and $w_i\bar{w}_i$ are each taken n -times
(MUST BE EXPLICIT)

For expanding w_i 's correctly and using the fact that

$$w_1 + w_2 + \dots + w_n = 0$$

and

$$\overline{w_1 + w_2 + \dots + w_n} = 0$$

$$\text{or } \bar{w}_1 + \bar{w}_2 + \dots + \bar{w}_n = 0$$